

DATABASE MANAGEMENT SYSTEM

DBMS ASSIGNMENT

CAB BOOKING SYSTEM

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INTRODUCTION:

Car Booking System is a database based system which enables customers to book cars to travel from one place to other places. This system maintains and manages vital information such as customer information, car information, driver information, bookings and payments.

This system makes use of database to maintain and organize all data and makes it easy to access. The use of primary key and foreign key will be used in this system.

In general, Car Booking System will be useful in managing car booking and other relevant information.

Name and Purpose of the Application:

Application Name: Car Booking System

Purpose:

A Car Booking System is a database management system where customers can reserve their cars for transportation purposes.

The system stores data related to customers, drivers, cars, reservations and payments. This system assists in keeping track of which customer reserved the car, which driver will take him, and how much he paid for the reservation.

Required Table:

1.	Customer
2.	Driver
3.	Car
4.	Bookings
5.	Payments

Column/Attributes for Each Table:

Customer Table:

Column	Description	Key
Customer_ID	Unique ID of customer	Primary Key
Customer_Name	Name of customer	-
Phone	Customer phone number	-
Email	Customer email	-
Address	Customer address	-

Driver Table:

Column	Description	Key
Driver_ID	Unique ID of driver	Primary Key
Driver_Name	Name of driver	—
Phone	Driver phone number	—
License_No	Driving license number	—
Availability	Driver availability status	—

Car Table:

Column	Description	Key
Car_ID	Unique ID of car	Primary Key
Car_Number	Vehicle registration number	—
Car_Model	Model of car	—
Car_Type	Sedan/SUV/Hatchback etc.	—
Fare_Per_KM	Cost per kilometre	—
Driver_ID	Driver assigned to car	Foreign Key

Booking Table:

Column	Description	Key
Booking_ID	Unique booking ID	Primary Key
Customer_ID	Customer making booking	Foreign Key
Car_ID	Car booked	Foreign Key
Booking_Date	Date of booking	—
Pickup_Location	Starting location	—
Drop_Location	Destination	—
Distance_KM	Travel distance	—
Booking_Status	Booking status	—

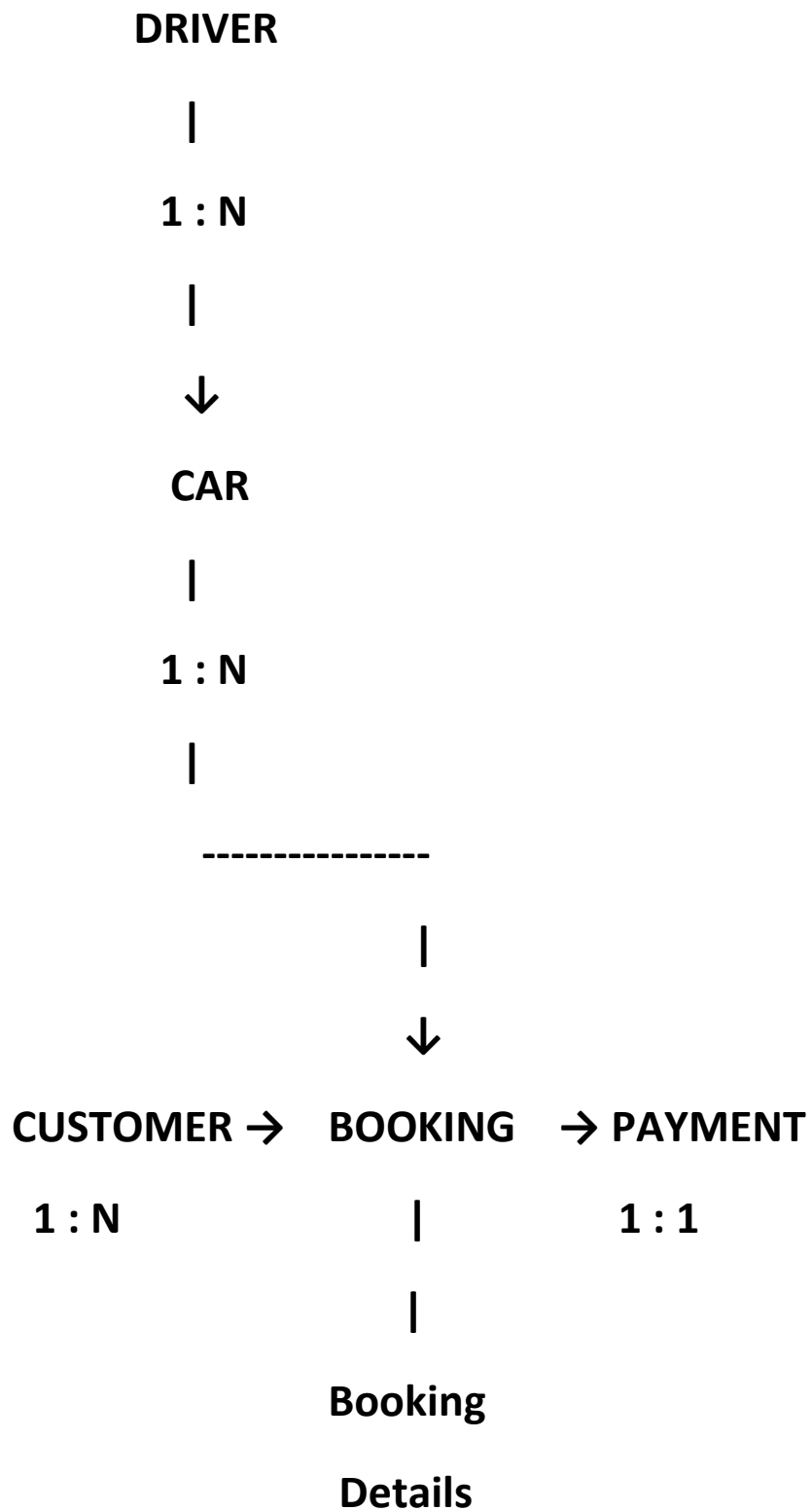
Payment Table:

Column	Description	Key
Payment_ID	Unique payment ID	Primary Key
Booking_ID	Booking associated with payment	Foreign Key
Amount	Amount paid	—
Payment_Date	Date of payment	—
Payment_Method	Cash/UPI/Card	—
Payment_Status	Paid/Pending	—

Relationship Between Tables:

No.	Table 1	Relationship	Table 2	Type	Foreign Key
1	Customer	Makes	Booking	1 : N	Customer_ID
2	Driver	Is assigned to	Car	1 : N	Driver_ID
3	Car	Is used for	Booking	1 : N	Car_ID
4	Booking	Has	Payment	1 : 1	Booking_ID

ER Diagram:



Example:

Customer Table:

Customer_ID	Customer_Name	Phone	Email	Address
CU01	Rahul	9876543210	rahul@mail.com	Bangalore
CU02	Arjun	9876543211	arjun@mail.com	Chennai
CU03	Kiran	9876543212	kiran@mail.com	Krishnagiri

Driver Table:

Driver_ID	Driver_Name	Phone	License_No	Availability
D01	Ravi	9123456780	DL1001	Available
D02	Kumar	9123456781	DL1002	Busy
D03	Suresh	9123456782	DL1003	Available

Car Table:

Car_ID	Car_Number	Car_Model	Car_Type	Fare_Per_KM	Driver_ID
C01	KA01AB1234	Swift	Hatchback	₹12	D01
C02	KA02CD5678	City	Sedan	₹15	D02
C03	KA03EF9012	Creta	SUV	₹20	D03

Booking Table:

Booking_ID	Customer_ID	Car_ID	Booking_Date	Pickup	Drop	Distance	Status
B01	CU01	C01	07-09-2026	Airport	MG Road	25 km	Completed
B02	CU02	C02	07-09-2026	Mall	Railway Station	12 km	Completed
B03	CU03	C03	08-09-2026	College	Airport	30 km	Confirmed

Payment Table:

Payment_ID	Booking_ID	Amount	Payment_Date	Payment_Method	Status
P01	B01	₹300	07-09-2026	UPI	Paid
P02	B02	₹180	07-09-2026	Card	Paid
P03	B03	₹600	08-09-2026	UPI	Pending

Brief explanation of the purpose of each table:

No.	Table Name	Purpose
1	Customer	Stores the details of customers, such as Customer ID, name, phone number, email, and address.
2	Driver	Stores information about drivers, including Driver ID, name, phone number, license number, and availability.
3	Car	Stores details about cars, such as Car ID, registration number, model, type, fare per kilometre, and assigned driver.
4	Booking	Stores information about each booking, including customer, car, booking date, pickup location, drop location, distance, and booking status.
5	Payment	Stores payment details related to bookings, such as payment ID, booking ID, amount, payment date, payment method, and payment status.

Conclusion :

Car Booking System is an effective tool for managing customers, drivers, cars, bookings, and payments. Through the use of related database tables, the Car Booking System organizes the data in a well-structured manner. The primary key identifies the records in the table uniquely while the foreign key relates different tables.

